Deep Learning Project Analysis

Loss Functions and Evaluation Metrics for AI Interview Evaluator

A comprehensive study on optimizing multimodal AI systems combining NLP, Speech Processing, Computer Vision, and ANN.

Based on "A Comprehensive Survey of Loss Functions and Meirics in Deep Learning"

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Project Overview

The AI Interview Evaluator is a sophisticated multimodal deep learning system designed to automate candidate assessment in mock interviews.

By analyzing synchronized streams of audio, text, and video, the system provides a holistic evaluation of candidate performance that goes beyond simple keyword matching.

System Outputs

- Emotion Analysis
- Confidence Score
- Response Relevance
- Final Score

Speech Recognition

Utilizes OpenAI's Whisper model to convert spoken candidate responses into accurate text transcripts for analysis.

Natural Language Processing

BERT-based models analyze semantic relevance, coherence, and sentiment of the textual responses.

Computer Vision

CNN models detect micro-expressions and facial emotions to gauge candidate composure and stress levels.

Audio Analysis & ANN

Extracts MFCCs, pitch, and speech rate features. A Neural Network fuses all multimodal inputs for final scoring.

AI Tasks in Our System

The AI Interview Evaluator integrates multiple machine learning paradigms

CLASSIFICATION

Predicting categorical labels from a fixed set of classes.

- **Emotion Detection:** Categorizing facial expressions (Happy, Neutral, Stressed).
- **Confidence Scoring:** Classifying vocal cues (Confident, Average, Nervous).
- **Sentiment Analysis**: Determining polarity of candidate responses.

REGRESSION

Predicting continuous numerical values based on input features.

- **Final Performance Score:** The ANN model fuses multimodal features to output a precise score (0-100).
- **Feature Integration:** Combines probabilities and confidence metrics into a single value.

SPEECH RECOGNITION

Converting spoken language into written text for analysis.

- **Whisper Model:** Transcribes candidate answers from audio to text.
- **Evaluation:** Performance is measured by transcription accuracy rather than loss.

Note: Each task type requires specific loss functions and evaluation metrics to accurately measure and optimize model performance.

What is a Loss Function?

Definition

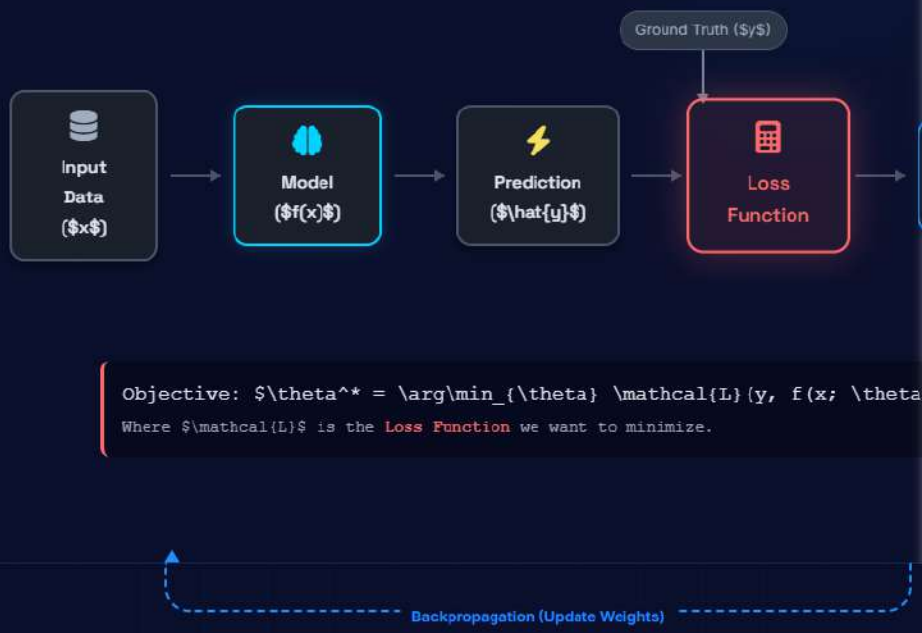
A loss function is a mathematical algorithm that quantifies the error between a model's **predictions** and the actual **ground truth**. It serves as the guide for the neural network during training.

→ Goal: Minimize Loss → Maximize Accuracy

Application in Our System

- **Classification:** Cross-Entropy & Focal Loss (Emotion, Sentiment)
- **Regression:** MSE & MAE (Interview Score)

THE TRAINING LOOP



Objective: $\theta^* = \arg\min_{\theta} \mathcal{L}(y, f(x; \theta

Where $\mathcal{L}$ is the **Loss Function** we want to minimize.

Cross Entropy Loss

The standard objective function for multi-class classification tasks

MATHEMATICAL FORMULATION

$$L_{CE} = -\sum_{c=1}^{M} y_c \log(p_c)$$

Where:

M = Number of classes (e.g., Happy, Neutral, Nervous)

y_c = Binary indicator (0 or 1) if class label *c* is the correct classification

p_c = Predicted probability for class *c* (output of Softmax)

REAL-WORLD EXAMPLE: EMOTION DETECTION

Target: **Happy** (Class 1)

Happy	0.85
Neutral	0.10
Nervous	0.05

Result: Low loss because the model assigned high probability to the correct class.

Advantages

- **Probabilistic Interpretation:** Directly maximizes the likelihood of the correct class.
- **Stable Gradients:** Works exceptionally well with Softmax, preventing vanishing gradients during backpropagation.
- **Standard:** Default choice for almost all deep learning classification tasks.

Limitations

- **Class Imbalance Sensitivity:** Dominated by easy/frequent examples, leading to poor performance on rare classes.
- **Outlier Sensitivity:** Penalizes incorrect confident predictions heavily, which can be problematic with noisy labels.

Focal Loss

Handling Class Imbalance in Emotion Detection

MATHEMATICAL FORMULATION

$$FL(p_t) = -\alpha_t (1 - p_t)^{\gamma} \log(p_t)$$

Where:
p_t = Probability of the true class (ground truth)
γ (Gamma) = Focusing parameter (usually ≥ 0). Reduces loss for easy examples.
α (Alpha) = Balancing factor to weight positive/negative classes.

IMPACT OF FOCUSING ($\Gamma = 2$)

Easy Example ($p_t = 0.9$)

Cross Entropy — High

Focal Loss — ~0 (ignored)

Hard Example ($p_t = 0.1$)

Cross Entropy

Focal Loss — High (Focus)

Focal Loss effectively "down-weights" easy examples to focus training on hard ones.

Advantages

- **Handles Imbalance:** Specifically designed for datasets where one class dominates (e.g., Neutral vs. Stress).
- **Focuses on Hard Samples:** The modulating factor $(1-p_t)^{\gamma}$ reduces the loss contribution from easy, well-classified examples.
- **Improves Minority Recall:** Helps the model learn rare emotions like "Anxiety" or "Excitement" better.

Limitations

- **Hyperparameter Tuning:** Requires careful tuning of α (alpha) and γ (gamma) for optimal performance.
- **Slower Convergence:** May take longer to train compared to standard Cross Entropy due to loss scaling.
- **Complexity:** More complex to implement and debug than standard loss functions.

Evaluating Model Performance

Classification Metrics: Confusion Matrix

Fundamental Tool for Classification

PREDICTED CLASS

ACTUAL CLASS	POSITIVE (1)	NEGATIVE (0)
POSITIVE (1)	**TP** True Positive — Correctly identified emotion/intent	**FN** False Negative — Missed detection (Type II Error)
NEGATIVE (0)	**FP** False Positive — False alarm (Type I Error)	**TN** True Negative — Correctly rejected

Accuracy

(TP + TN) / Total

Overall correctness of the model. Best for balanced datasets.

Precision

TP / (TP + FP)

"Of all positive predictions, how many were actually correct?"

Important when False Positives are costly.

Recall (Sensitivity)

TP / (TP + FN)

"Of all actual positives, how many did we find?"

Important when missing a positive is dangerous.

Multi-Class Averaging Methods

Macro Avg
Arithmetic mean of metrics for each class. Treats all classes equally.

Weighted Avg
Mean weighted by the number of true instances (support) for each class.

Micro Avg
Aggregates global TP, FP, FN counts. Equal to accuracy in multi-class.

Accuracy & Precision

Fundamental metrics for evaluating classification models

Accuracy

(TP + TN) / Total Predictions

Measures the proportion of correct predictions (both true positives and true negatives) out of all predictions made by the model.

ADVANTAGES

- ✓ Simple to understand and interpret as a single percentage.
- ✓ Effective when classes are balanced (e.g., 50/50 split).

LIMITATIONS

- × Highly misleading for imbalanced datasets.
- ▲ **Example:** In a dataset with 90% "Neutral" emotions, a model predicting only "Neutral" gets 90% accuracy but is useless.

Precision

TP / (TP + FP)

Measures the quality of the positive predictions. Out of all instances predicted as positive, how many were actually positive?

ADVANTAGES

- ✓ Crucial when False Positives have a high cost.
- ✓ **Example:** If the model labels a candidate as "Confident", we want to be sure they actually are.

LIMITATIONS

- × Ignores False Negatives (missed detections).
- ▲ A model can have 100% precision by predicting only one correct instance and missing all others.

Recall & F1 Score

Metrics for sensitivity and balanced performance evaluation

Recall

$$TP / (TP + FN)$$

Measures the ability of the model to find all the relevant cases within a dataset. Also known as Sensitivity or True Positive Rate.

BEST FOR

- ✓ Situations where missing a positive case is costly (e.g., detecting fraud or disease).
- ✓ **Example:** Identifying every "Confident" candidate so none are overlooked.

LIMITATIONS

- ✗ Can be easily maximized by predicting everything as positive.
- ⚠ Improving recall often leads to more False Positives (lower precision).

F1 Score

$$2 \times (Precision \times Recall) / (Precision + Recall)$$

The harmonic mean of Precision and Recall. It provides a single metric that balances both concerns.

BEST FOR

- ✓ Imbalanced datasets where you need a balance between precision and recall.
- ✓ **Example:** Comparing overall performance of different emotion detection models.

LIMITATIONS

- ✗ Less intuitive to interpret than accuracy.
- ⚠ Treats precision and recall as equally important, which may not always be desired.

Regression Loss Functions

Predicting Continuous Numerical Values

Project Component: Final Scoring

Definition

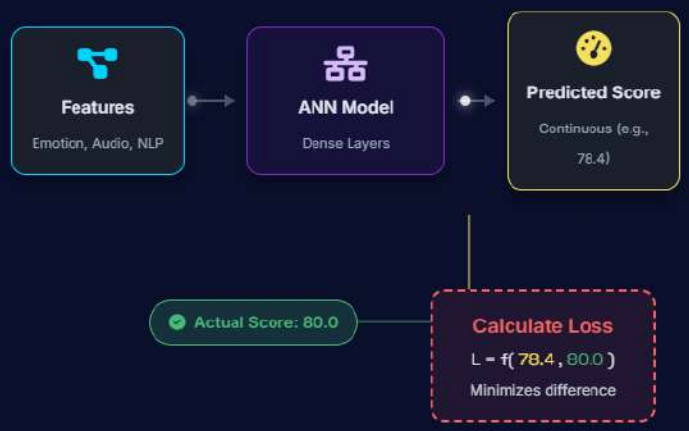
Regression loss functions measure the distance between a predicted continuous value (scalar) and the actual target value. The goal is to minimize this distance.

Project Application

The ANN model predicts a final **Interview Performance Score** (0-100) based on multimodal features.

MSE	Standard choice; penalizes large errors.
MAE	Robust to outliers in scoring data.
Huber	Balanced approach (Best of both).

REGRESSION PIPELINE



Mean Squared Error (MSE)

The standard loss function for regression tasks like score prediction

MATHEMATICAL FORMULATION

$$MSE = 1/n \sum (y_i - \hat{y}_i)^2$$

Where:

- n = Total number of samples in the batch
- y_i = Actual Ground Truth value (e.g., Interview Score 85)
- $\hat{y}_i$ = Model's Predicted value (e.g., Interview Score 60)

IMPACT OF SQUARING ERRORS

Target: 85 | Pred: 60 (Diff: 25) — Loss: 625

Large deviations are heavily penalized.

Target: 85 | Pred: 82 (Diff: 3) — Loss: 9

Small errors result in negligible loss.

Advantages

- **Strong Penalty:** Squaring the error ensures that large mistakes are prioritized during training, forcing the model to correct them quickly.
- **Convexity:** It is a smooth, differentiable, and convex function, making it easy to optimize using Gradient Descent.
- **Mathematical Convenience:** The gradient is linear (2 * error), providing stable updates near the minimum.

Limitations

- **Sensitive to Outliers:** A single anomalous data point can disproportionately skew the loss, pulling the model away from the general trend.
- **Unit Mismatch:** The loss is in "squared units" (e.g., score squared), which makes it less interpretable than Mean Absolute Error.

Mean Absolute Error (MAE)

A robust L1 loss function for regression tasks

MATHEMATICAL FORMULATION

$$MAE = 1/n \sum_{i=1}^{n} |y_i - \hat{y}_i|$$

Where:

- n = Total number of samples
- y_i = Actual value (Ground Truth)
- $\hat{y}_i$ = Predicted value by the model
- $|...|$ = Absolute difference (L1 Norm)

ROBUSTNESS TO OUTLIERS EXAMPLE

Scenario: An outlier prediction error of 50 points

MAE Penalty **50**	Linear Growth $\|50\| = 50$
MSE Penalty (Reference) **2,500**	Quadratic Growth $50^2 = 2500$

Result: MAE is much less sensitive to outliers than MSE.

Advantages

- **Robust to Outliers:** Does not heavily penalize large errors, making it stable for noisy data.
- **Interpretable Units:** The error is in the same units as the target variable (e.g., Interview Score points).
- **Median Estimation:** Theoretically estimates the median of the conditional distribution.

Limitations

- **Unstable Gradients:** The gradient is constant (±1), which can cause the optimizer to overshoot the minimum.
- **Non-Differentiable at 0:** Requires sub-gradients or smoothing (like Huber Loss) for efficient training close to zero error.

Robust Regression

Huber Loss

Hybrid of MSE & MAE

PIECEWISE DEFINITION

$$L_\delta(a) = \begin{cases} \frac{1}{2} a^2 & \text{for } |a| \le \delta \\ \delta\left(|a| - \frac{1}{2}\delta\right) & \text{otherwise} \end{cases}$$

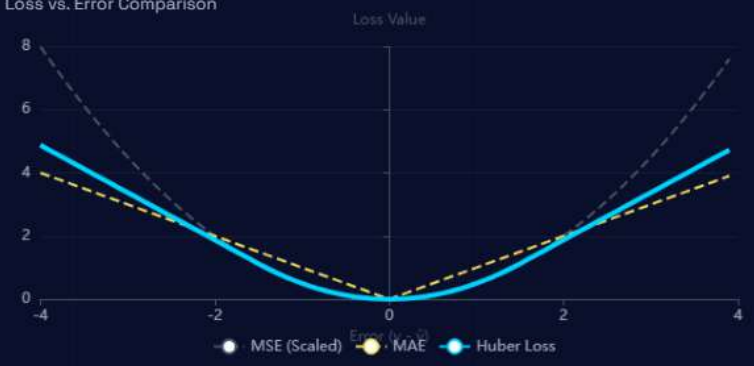
Where $a = y - \hat{y}$ (residual) and δ is the threshold.

The Threshold Parameter (δ)

The delta (δ) parameter dictates where the loss function transitions from quadratic (MSE-like) to linear (MAE-like).

Small Errors (|a| ≤ δ)	Large Errors (|a| > δ)
Quadratic	**Linear**

Loss vs. Error Comparison



Advantages

- **Robust to Outliers:** Doesn't penalize large errors as heavily as MSE (linear tail).
- **Differentiable at 0:** Smooth curve at the origin, unlike MAE.

Limitations

- **Hyperparameter Tuning:** Requires careful selection of δ for optimal performance.
- Computational complexity is slightly higher than pure MSE/MAE.

Regression Evaluation Metrics

Performance measurement for the Final Interview Score prediction (ANN Model)

RMSE

Root Mean Squared Error

$$\sqrt{\left[(1/n) \Sigma (y - \hat{y})^2 \right]}$$

Measures the square root of the average squared difference between predicted and actual values. It represents the standard deviation of the prediction errors.

ADVANTAGES

- ✓ Easy to interpret: Errors are in the same units as the interview score (e.g., points).
- ✓ Strongly penalizes large prediction errors, which is useful if big mistakes are unacceptable.

LIMITATIONS

- × Sensitive to outliers; a few very bad predictions can skew the score significantly.

R^2 Score

Coefficient of Determination

$$1 - (SS_{res} / SS_{tot})$$

Indicates the proportion of the variance in the dependent variable (Interview Score) that is predictable from the independent variables (multimodal features).

ADVANTAGES

- ✓ Scale-independent (0 to 1), making it easy to compare model fit across different datasets.
- ✓ Provides a clear measure of "Goodness of Fit" relative to a simple mean baseline.

LIMITATIONS

- × Does not indicate the magnitude of the error (how far off the score actually is).

Speech Recognition Evaluation

Word Error Rate (WER)

Whisper Model Evaluation

What is WER?

The standard metric for Automatic Speech Recognition (ASR). It measures the edit distance between the reference transcript and the model's output, normalized by the length of the reference.

$$WER = (S + D + I) / N$$

S **Substitutions**
Wrong word predicted

D **Deletions**
Word missed

I **Insertions**
Extra word added

N **Total Words**
In reference text

Transcription Example

REFERENCE: I am confident about this position

PREDICTED: I am confidence (SUB) about this position

S=1 + D=0 + I=0 | Total Words (N) = 6 | **WER = 16.7%**

Performance Interpretation

0% (Perfect) | 10-20% (Acceptable) | >30% (Poor)

Lower WER indicates better speech recognition accuracy.

System Architecture Analysis

Comparison of Loss Functions

Selected for Optimal Performance

LOSS FUNCTION	TASK TYPE	USED IN PROJECT	ADVANTAGES	LIMITATIONS
Cross Entropy	CLASSIFICATION	Emotion detection, Sentiment analysis, Confidence	✓ Standard for multi-class ✓ Probabilistic output	× Sensitive to class imbalance
Focal Loss	CLASSIFICATION	Rare emotions (e.g., Stress, Anxiety detection)	✓ Focuses on hard samples ✓ Handles extreme imbalance	× Requires hyperparameter tuning
Mean Squared Error	REGRESSION	Final Interview Performance Score Prediction	✓ Penalizes large errors strongly ✓ Convex & differentiable	× Very sensitive to outliers
Mean Absolute Error	REGRESSION	Alternative for robust scoring model	✓ Robust to outliers ✓ Linear penalty	× Non-differentiable at 0 × Slower convergence
Huber Loss	REGRESSION	Balanced scoring (Hybrid approach)	✓ Best of MSE & MAE ✓ Robust & smooth	× Requires threshold (δ) selection

● Classification Tasks (Categorical) ● Regression Tasks (Continuous)

System Performance Analysis

Comparison of Evaluation Metrics

Classification • Regression • Speech

EVALUATION METRIC	TASK TYPE	USED IN PROJECT	ADVANTAGES	LIMITATIONS
Accuracy	CLASSIFICATION	Emotion detection, Sentiment analysis	✓ Simple and intuitive ✓ Easy to interpret	× Misleading on imbalanced datasets
Precision	CLASSIFICATION	Confidence detection (Avoid false positives)	✓ Measures correctness of positives ✓ Minimizes costly false alarms	× Ignores missed positive cases
Recall	CLASSIFICATION	Sentiment / Emotion detection	✓ Captures all relevant cases ✓ Good for critical detection	× May increase false positives
F1 Score	CLASSIFICATION	General Emotion detection quality	✓ Balances precision and recall ✓ Robust for imbalance	× Less intuitive to interpret
RMSE	REGRESSION	Final Interview Score Prediction	✓ Error in same units as output ✓ Penalizes large errors	× Sensitive to outliers
R^2 Score	REGRESSION	ANN Scoring Model Fit	✓ Shows goodness of fit (0-1) ✓ Standard comparison metric	× Doesn't show error magnitude

● Classification Tasks ● Regression Tasks ● Speech Recognition

Conclusion & Key Takeaways

System Architecture Final Summary

Project Completed

Our AI Interview Evaluator system effectively combines multimodal inputs. We have identified and selected the optimal loss functions and metrics for each component to ensure reliable performance assessment.

Classification

Emotion, Sentiment, Confidence

- Objective: **Cross-Entropy Loss**
- Imbalance: **Focal Loss**
- Metrics: **F1 Score & Accuracy**

Regression

Final Performance Scoring

- Standard: **MSE** (Mean Squared Error)
- Robustness: **Huber Loss**
- Metrics: **RMSE & R^2 Score**

Speech Recog.

Whisper Audio Transcription

- Model: **OpenAI Whisper**
- Eval Metric: **WER** (Word Error Rate)
- Goal: **Minimizing Substitutions**

Future Directions

We will focus on fine-tuning hyperparameters (γ for Focal Loss, δ for Huber) and implementing data augmentation to further improve robustness.

THANK YOU

Questions & Discussion